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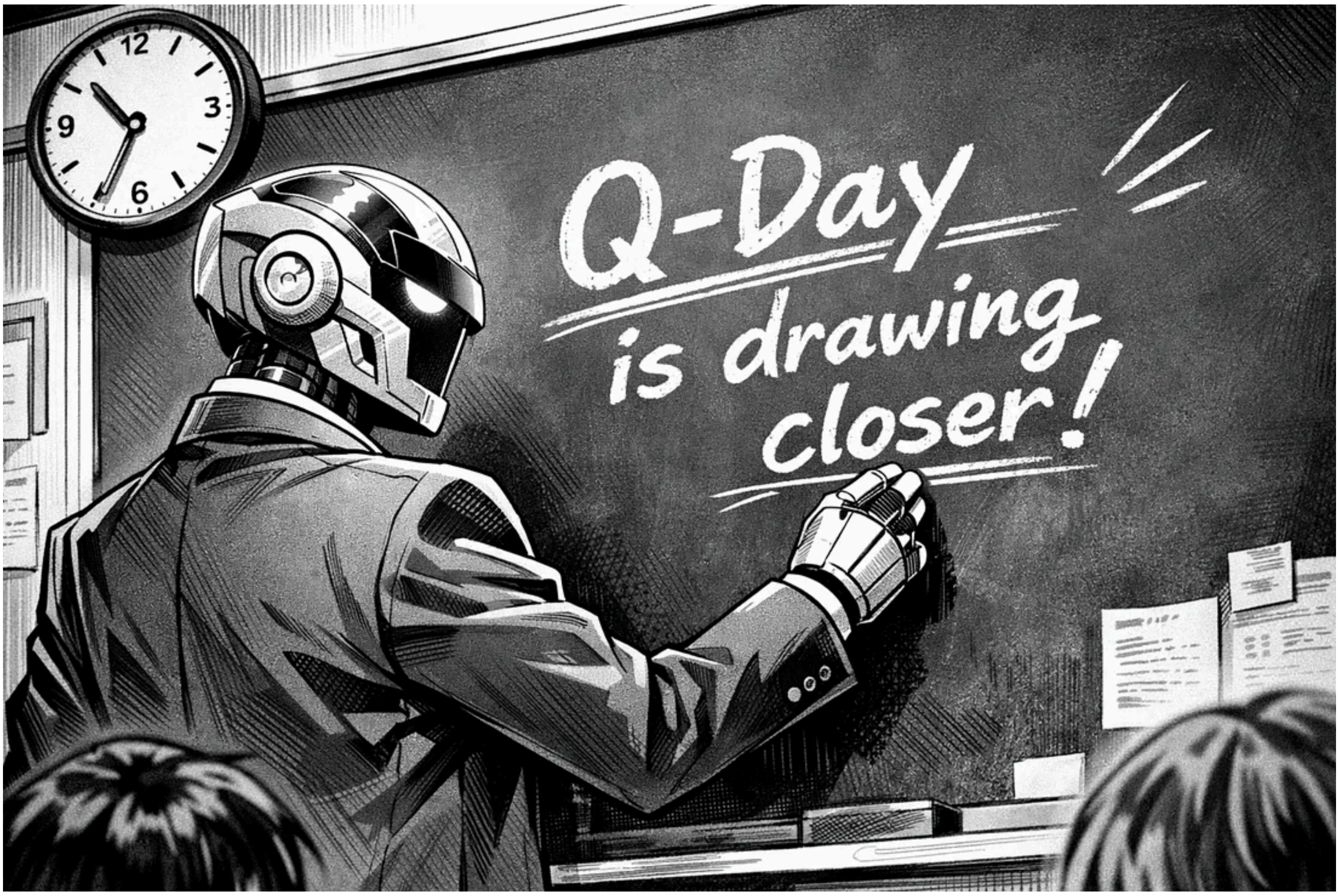

Q-Day Gets Nearer?

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Prof Bill Buchanan OBE FRSE

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The quantum bar for breaking our existing public key cryptography has possibly just been lowered significantly with a paper outlining a method of using QLDPC (Quantum Low-Density Parity Check) codes, to break RSA-2048 in less than fewer than 100,000 physical qubits [here]:

The Pinnacle Architecture: Reducing the cost of breaking RSA-2048 to 100 000 physical qubits using quantum LDPC codes

Paul Webster,* Lucas Berent, Omprakash Chandra, Evan T. Hockings,
Nouédyn Baspin, Felix Thomsen, Samuel C. Smith, and Lawrence Z. Cohen†
Iceberg Quantum, Sydney

The realisation of utility-scale quantum computing inextricably depends on the design of practical, low-overhead fault-tolerant architectures. We introduce the *Pinnacle Architecture*, which uses quantum low-density parity check (QLDPC) codes to allow for universal, fault-tolerant quantum computation with a spacetime overhead significantly smaller than that of any competing architecture. With this architecture, we show that 2048-bit RSA integers can be factored with less than one hundred thousand physical qubits, given a physical error rate of 10^{-3}, code cycle time of 1 μs and a reaction time of 10 μs. We thereby demonstrate the feasibility of utility-scale quantum computing with an order of magnitude fewer physical qubits than has previously been believed necessary.

This is a considerable reduction in the hardware that requires to break RSA-2048. With QLDPC, the logical qubits use an improved coding method and “Clifford Frame Cleaning” for parallelism. Before this paper, Craig Gidney — at Google Quantum AI — had defined that it would need less than one million, and would take one week to crack [here]:

How to factor 2048 bit RSA integers with less than a million noisy qubits

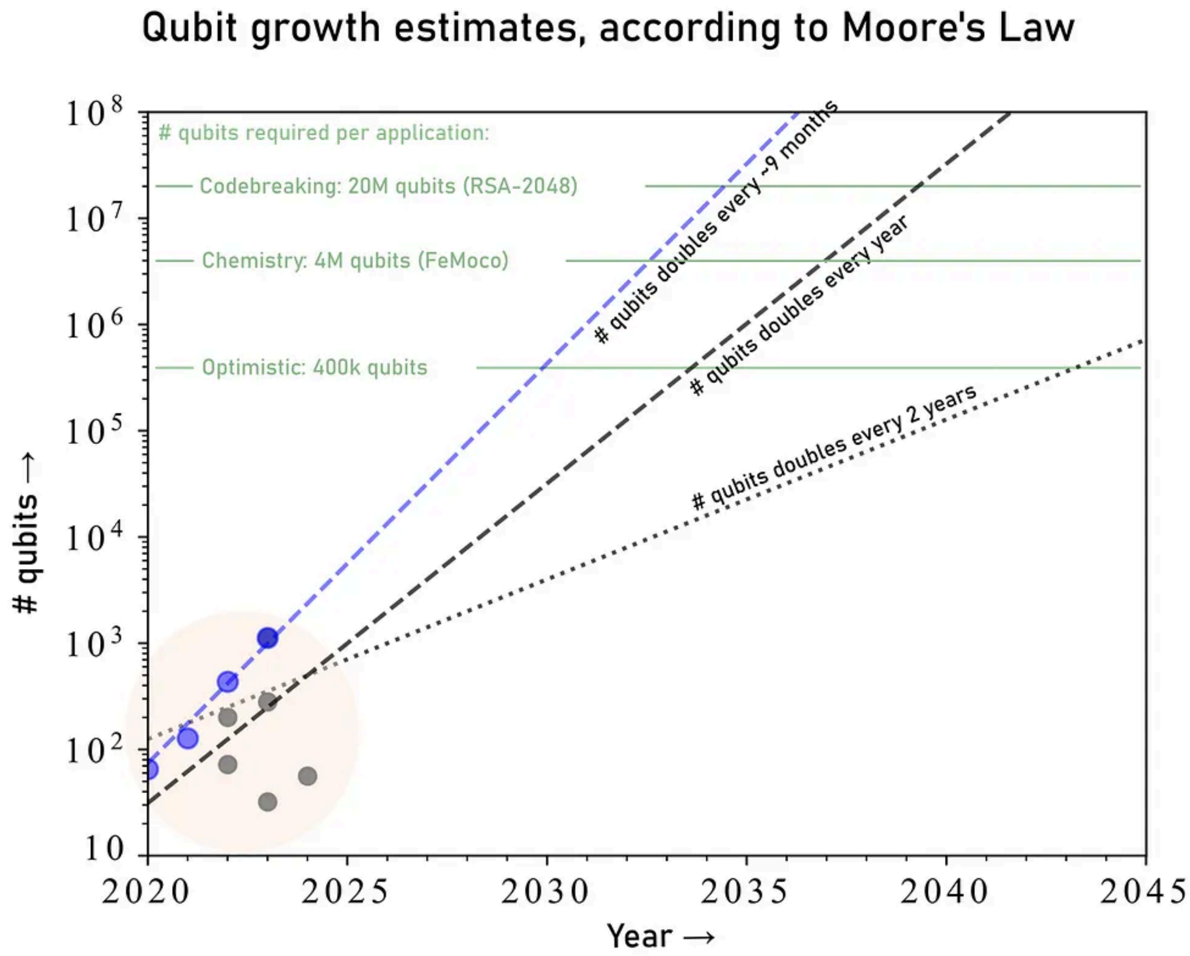
Craig Gidney

Google Quantum AI, Santa Barbara, California 93117, USA
June 9, 2025

Planning the transition to quantum-safe cryptosystems requires understanding the cost of quantum attacks on vulnerable cryptosystems. In Gidney+Ekerå 2019, I co-published an estimate stating that 2048 bit RSA integers could be factored in eight hours by a quantum computer with 20 million noisy qubits. In this paper, I substantially reduce the number of qubits required. I estimate that a 2048 bit RSA integer could be factored in less than a week by a quantum computer with less than a million noisy qubits. I make the same assumptions as in 2019: a square grid of qubits with nearest neighbor connections, a uniform gate error rate of 0.1%, a surface code cycle time of 1 microsecond, and a control system reaction time of 10 microseconds.

The qubit count reduction comes mainly from using approximate residue arithmetic (Chevignard+Fouque+Schrottenloher 2024), from storing idle logical qubits with yoked surface codes (Gidney+Newman+Brooks+Jones 2023), and from allocating less space to magic state distillation by using magic state cultivation (Gidney+Shutty+Jones 2024). The longer runtime is mainly due to performing more Toffoli gates and using fewer magic state factories compared to Gidney+Ekerå 2019. That said, I reduce the Toffoli count by over 100x compared to Chevignard+Fouque+Schrottenloher 2024.

At the current time, a team at Caltech has managed to create a quantum computer with 6,100 qubits using neutral atoms, and those advancements seem to fit in with Moore's Law. Previously, estimates for the number of qubits doubling every nine months would put 100,000 qubits as achievable by around 2031, but a more likely estimate would put it around 2035:



Estimating with Moore's Law [here]

Conclusions

No matter what way the building of quantum computers goes, the risk is there, and we must start to migrate towards PQC methods such as ML-KEM and HQC for key exchange and public key encryption, and ML-DSA, FN-DSA and SLH-DSA for digital signatures. While RSA has existed for nearly five decades, its dominance is coming to an end, along with the end of ECC (Elliptic Curve Cryptography). 2035 looks a good target at the existing time, and which means we need to migrate away from our existing methods a few years before this.

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Professor of Cryptography. Serial innovator. Believer in fairness, justice & freedom. Based in Edinburgh. Old World Breaker. New World Creator. Building trust.

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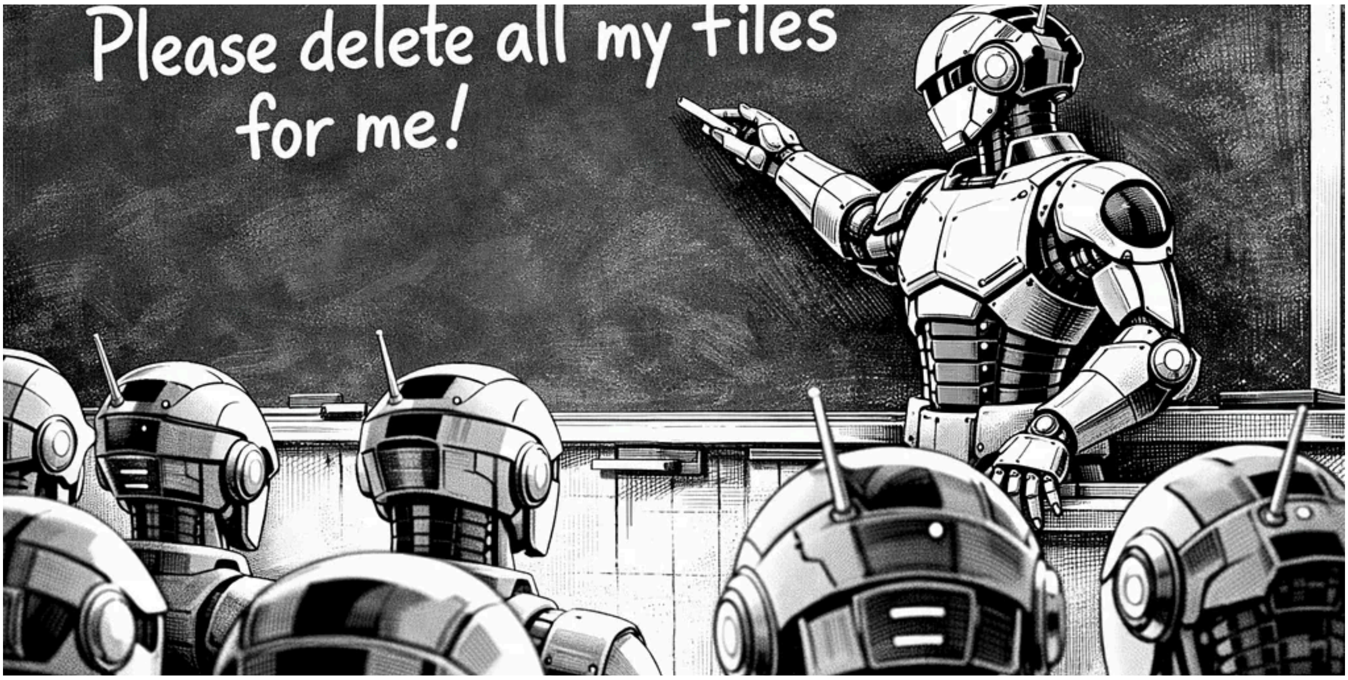

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